Introduction to linear regression

Guibril Ramde

The Human Freedom Index is a report that attempts to summarize the idea of "freedom" through a bunch of different variables for many countries around the globe. It serves as a rough objective measure for the relationships between the different types of freedom - whether it's political, religious, economical or personal freedom - and other social and economic circumstances. The Human Freedom Index is an annually co-published report by the Cato Institute, the Fraser Institute, and the Liberales Institut at the Friedrich Naumann Foundation for Freedom.

In this lab, you'll be analyzing data from Human Freedom Index reports from 2008-2016. Your aim will be to summarize a few of the relationships within the data both graphically and numerically in order to find which variables can help tell a story about freedom.

Getting Started

Load packages

In this lab, you will explore and visualize the data using the **tidyverse** suite of packages. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let's load the packages.

```
library(tidyverse)
library(openintro)
data('hfi', package='openintro')
```

The data

The data we're working with is in the openintro package and it's called `hfi`, short for Human Freedom Index.

1. What are the dimensions of the dataset?

```
dim(hfi)
```

```
## [1] 1458  123
```

```
glimpse(hfi)
```

```
## Rows: 1,458
## Columns: 123
## $ year                              <dbl> 2016, 2016, 2016, 2016, 2016, 2016,~
## $ ISO_code                          <chr> "ALB", "DZA", "AGO", "ARG", "ARM", ~
## $ countries                         <chr> "Albania", "Algeria", "Angola", "Ar~
```

```
## $ region                            <chr> "Eastern Europe", "Middle East & No~
## $ pf_rol_procedural                 <dbl> 6.661503, NA, NA, 7.098483, NA, 8.4~
## $ pf_rol_civil                      <dbl> 4.547244, NA, NA, 5.791960, NA, 7.5~
## $ pf_rol_criminal                   <dbl> 4.666508, NA, NA, 4.343930, NA, 7.3~
## $ pf_rol                            <dbl> 5.291752, 3.819566, 3.451814, 5.744~
## $ pf_ss_homicide                    <dbl> 8.920429, 9.456254, 8.060260, 7.622~
## $ pf_ss_disappearances_disap        <dbl> 10, 10, 5, 10, 10, 10, 10, 10, 10, ~
## $ pf_ss_disappearances_violent      <dbl> 10.000000, 9.294030, 10.000000, 10.~
## $ pf_ss_disappearances_organized    <dbl> 10.0, 5.0, 7.5, 7.5, 7.5, 10.0, 10.~
## $ pf_ss_disappearances_fatalities   <dbl> 10.000000, 9.926119, 10.000000, 10.~
## $ pf_ss_disappearances_injuries     <dbl> 10.000000, 9.990149, 10.000000, 9.9~
## $ pf_ss_disappearances              <dbl> 10.000000, 8.842060, 8.500000, 9.49~
## $ pf_ss_women_fgm                   <dbl> 10.0, 10.0, 10.0, 10.0, 10.0, 10.0,~
## $ pf_ss_women_missing               <dbl> 7.5, 7.5, 10.0, 10.0, 5.0, 10.0, 10~
## $ pf_ss_women_inheritance_widows    <dbl> 5, 0, 5, 10, 10, 10, 10, 5, NA, 0, ~
## $ pf_ss_women_inheritance_daughters <dbl> 5, 0, 5, 10, 10, 10, 10, 10, NA, 0,~
## $ pf_ss_women_inheritance           <dbl> 5.0, 0.0, 5.0, 10.0, 10.0, 10.0, 10~
## $ pf_ss_women                       <dbl> 7.500000, 5.833333, 8.333333, 10.00~
## $ pf_ss                             <dbl> 8.806810, 8.043882, 8.297865, 9.040~
## $ pf_movement_domestic              <dbl> 5, 5, 0, 10, 5, 10, 10, 5, 10, 10, ~
## $ pf_movement_foreign               <dbl> 10, 5, 5, 10, 5, 10, 10, 5, 10, 5, ~
## $ pf_movement_women                 <dbl> 5, 5, 10, 10, 10, 10, 10, 5, NA, 5,~
## $ pf_movement                       <dbl> 6.666667, 5.000000, 5.000000, 10.00~
## $ pf_religion_estop_establish       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pf_religion_estop_operate         <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pf_religion_estop                 <dbl> 10.0, 5.0, 10.0, 7.5, 5.0, 10.0, 10~
## $ pf_religion_harassment            <dbl> 9.566667, 6.873333, 8.904444, 9.037~
## $ pf_religion_restrictions          <dbl> 8.011111, 2.961111, 7.455556, 6.850~
## $ pf_religion                       <dbl> 9.192593, 4.944815, 8.786667, 7.795~
## $ pf_association_association        <dbl> 10.0, 5.0, 2.5, 7.5, 7.5, 10.0, 10.~
## $ pf_association_assembly           <dbl> 10.0, 5.0, 2.5, 10.0, 7.5, 10.0, 10~
## $ pf_association_political_establish <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pf_association_political_operate  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pf_association_political          <dbl> 10.0, 5.0, 2.5, 5.0, 5.0, 10.0, 10.~
## $ pf_association_prof_establish     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pf_association_prof_operate       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pf_association_prof               <dbl> 10.0, 5.0, 5.0, 7.5, 5.0, 10.0, 10.~
## $ pf_association_sport_establish    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pf_association_sport_operate      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pf_association_sport              <dbl> 10.0, 5.0, 7.5, 7.5, 7.5, 10.0, 10.~
## $ pf_association                    <dbl> 10.0, 5.0, 4.0, 7.5, 6.5, 10.0, 10.~
## $ pf_expression_killed              <dbl> 10.000000, 10.000000, 10.000000, 10~
## $ pf_expression_jailed              <dbl> 10.000000, 10.000000, 10.000000, 10~
## $ pf_expression_influence           <dbl> 5.0000000, 2.6666667, 2.6666667, 5.~
## $ pf_expression_control             <dbl> 5.25, 4.00, 2.50, 5.50, 4.25, 7.75,~
## $ pf_expression_cable               <dbl> 10.0, 10.0, 7.5, 10.0, 7.5, 10.0, 1~
## $ pf_expression_newspapers          <dbl> 10.0, 7.5, 5.0, 10.0, 7.5, 10.0, 10~
## $ pf_expression_internet            <dbl> 10.0, 7.5, 7.5, 10.0, 7.5, 10.0, 10~
## $ pf_expression                     <dbl> 8.607143, 7.380952, 6.452381, 8.738~
## $ pf_identity_legal                 <dbl> 0, NA, 10, 10, 7, 7, 10, 0, NA, NA,~
## $ pf_identity_parental_marriage     <dbl> 10, 0, 10, 10, 10, 10, 10, 10, 10, ~
## $ pf_identity_parental_divorce      <dbl> 10, 5, 10, 10, 10, 10, 10, 10, 10, ~
## $ pf_identity_parental              <dbl> 10.0, 2.5, 10.0, 10.0, 10.0, 10.0, ~
## $ pf_identity_sex_male              <dbl> 10, 0, 0, 10, 10, 10, 10, 10, 10, 1~
```

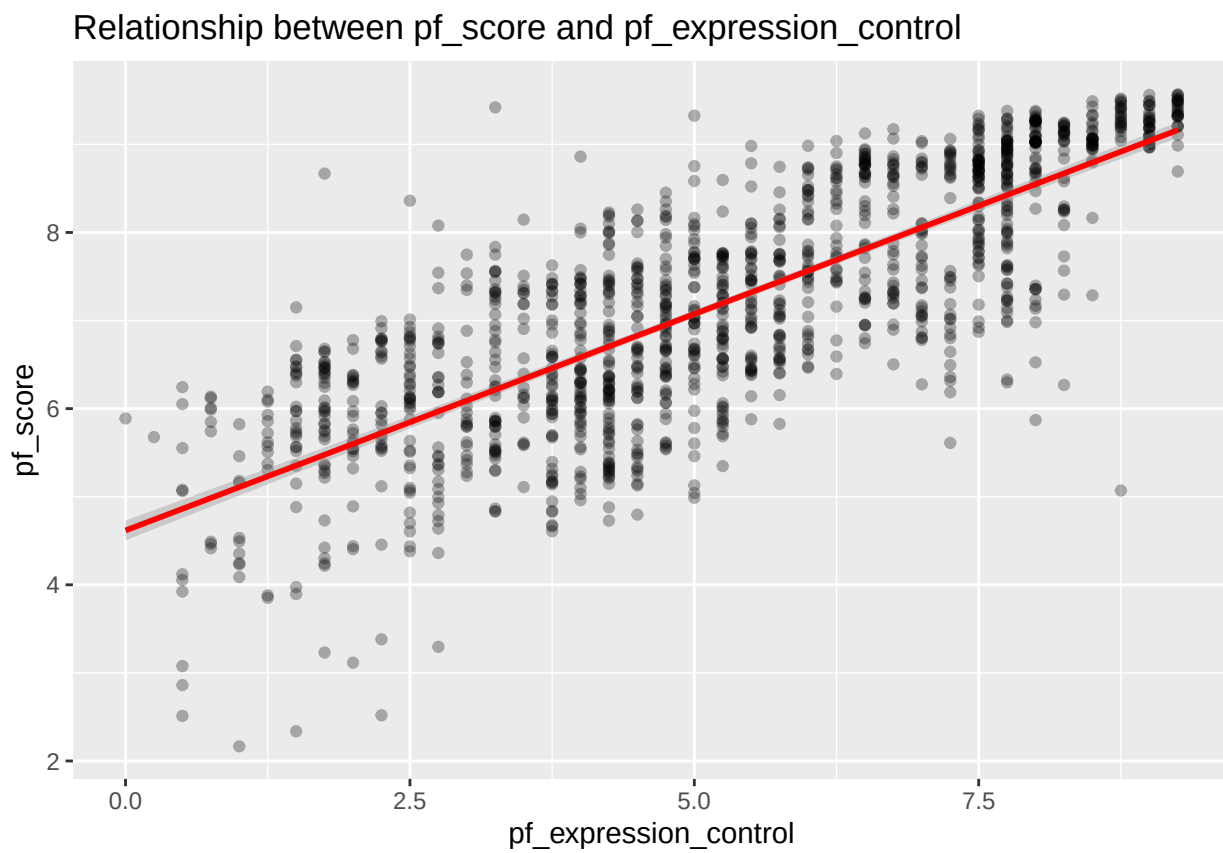
```
## $ pf_identity_sex_female            <dbl> 10, 0, 0, 10, 10, 10, 10, 10, 10, 1~
## $ pf_identity_sex                   <dbl> 10, 0, 0, 10, 10, 10, 10, 10, 10, 1~
## $ pf_identity_divorce               <dbl> 5, 0, 10, 10, 5, 10, 10, 5, NA, 0, ~
## $ pf_identity                       <dbl> 6.2500000, 0.8333333, 7.5000000, 10~
## $ pf_score                          <dbl> 7.596281, 5.281772, 6.111324, 8.099~
## $ pf_rank                           <dbl> 57, 147, 117, 42, 84, 11, 8, 131, 6~
## $ ef_government_consumption         <dbl> 8.232353, 2.150000, 7.600000, 5.335~
## $ ef_government_transfers           <dbl> 7.509902, 7.817129, 8.886739, 6.048~
## $ ef_government_enterprises         <dbl> 8, 0, 0, 6, 8, 10, 10, 0, 7, 10, 7,~
## $ ef_government_tax_income          <dbl> 9, 7, 10, 7, 5, 5, 4, 9, 10, 10, 8,~
## $ ef_government_tax_payroll         <dbl> 7, 2, 9, 1, 5, 5, 3, 4, 10, 10, 8, ~
## $ ef_government_tax                 <dbl> 8.0, 4.5, 9.5, 4.0, 5.0, 5.0, 3.5, ~
## $ ef_government                     <dbl> 7.935564, 3.616782, 6.496685, 5.346~
## $ ef_legal_judicial                 <dbl> 2.6682218, 4.1867042, 1.8431292, 3.~
## $ ef_legal_courts                   <dbl> 3.145462, 4.327113, 1.974566, 2.930~
## $ ef_legal_protection               <dbl> 4.512228, 4.689952, 2.512364, 4.255~
## $ ef_legal_military                 <dbl> 8.333333, 4.166667, 3.333333, 7.500~
## $ ef_legal_integrity                <dbl> 4.166667, 5.000000, 4.166667, 3.333~
## $ ef_legal_enforcement              <dbl> 4.3874441, 4.5075380, 2.3022004, 3.~
## $ ef_legal_restrictions             <dbl> 6.485287, 6.626692, 5.455882, 6.857~
## $ ef_legal_police                   <dbl> 6.933500, 6.136845, 3.016104, 3.385~
## $ ef_legal_crime                    <dbl> 6.215401, 6.737383, 4.291197, 4.133~
## $ ef_legal_gender                   <dbl> 0.9487179, 0.8205128, 0.8461538, 0.~
## $ ef_legal                          <dbl> 5.071814, 4.690743, 2.963635, 3.904~
## $ ef_money_growth                   <dbl> 8.986454, 6.955962, 9.385679, 5.233~
## $ ef_money_sd                       <dbl> 9.484575, 8.339152, 4.986742, 5.224~
## $ ef_money_inflation                <dbl> 9.743600, 8.720460, 3.054000, 2.000~
## $ ef_money_currency                 <dbl> 10, 5, 5, 10, 10, 10, 10, 5, 0, 10,~
## $ ef_money                          <dbl> 9.553657, 7.253894, 5.606605, 5.614~
## $ ef_trade_tariffs_revenue          <dbl> 9.626667, 8.480000, 8.993333, 6.060~
## $ ef_trade_tariffs_mean             <dbl> 9.24, 6.22, 7.72, 7.26, 8.76, 9.50,~
## $ ef_trade_tariffs_sd               <dbl> 8.0240, 5.9176, 4.2544, 5.9448, 8.0~
## $ ef_trade_tariffs                  <dbl> 8.963556, 6.872533, 6.989244, 6.421~
## $ ef_trade_regulatory_nontariff     <dbl> 5.574481, 4.962589, 3.132738, 4.466~
## $ ef_trade_regulatory_compliance    <dbl> 9.4053278, 0.0000000, 0.9171598, 5.~
## $ ef_trade_regulatory               <dbl> 7.489905, 2.481294, 2.024949, 4.811~
## $ ef_trade_black                    <dbl> 10.00000, 5.56391, 10.00000, 0.0000~
## $ ef_trade_movement_foreign         <dbl> 6.306106, 3.664829, 2.946919, 5.358~
## $ ef_trade_movement_capital         <dbl> 4.6153846, 0.0000000, 3.0769231, 0.~
## $ ef_trade_movement_visit           <dbl> 8.2969231, 1.1062564, 0.1106256, 7.~
## $ ef_trade_movement                 <dbl> 6.406138, 1.590362, 2.044823, 4.697~
## $ ef_trade                          <dbl> 8.214900, 4.127025, 5.264754, 3.982~
## $ ef_regulation_credit_ownership    <dbl> 5, 0, 8, 5, 10, 10, 8, 5, 10, 10, 5~
## $ ef_regulation_credit_private      <dbl> 7.295687, 5.301526, 9.194715, 4.259~
## $ ef_regulation_credit_interest     <dbl> 9, 10, 4, 7, 10, 10, 10, 9, 10, 10,~
## $ ef_regulation_credit              <dbl> 7.098562, 5.100509, 7.064905, 5.419~
## $ ef_regulation_labor_minwage       <dbl> 5.566667, 5.566667, 8.900000, 2.766~
## $ ef_regulation_labor_firing        <dbl> 5.396399, 3.896912, 2.656198, 2.191~
## $ ef_regulation_labor_bargain       <dbl> 6.234861, 5.958321, 5.172987, 3.432~
## $ ef_regulation_labor_hours         <dbl> 8, 6, 4, 10, 10, 10, 6, 6, 8, 8, 10~
## $ ef_regulation_labor_dismissal     <dbl> 6.299741, 7.755176, 6.632764, 2.517~
## $ ef_regulation_labor_conscription  <dbl> 10, 1, 0, 10, 0, 10, 3, 1, 10, 10, ~
## $ ef_regulation_labor               <dbl> 6.916278, 5.029513, 4.560325, 5.151~
## $ ef_regulation_business_adm        <dbl> 6.072172, 3.722341, 2.758428, 2.404~
```

```
## $ ef_regulation_business_bureaucracy <dbl> 6.000000, 1.777778, 1.333333, 6.666~
## $ ef_regulation_business_start       <dbl> 9.713864, 9.243070, 8.664627, 9.122~
## $ ef_regulation_business_bribes      <dbl> 4.050196, 3.765515, 1.945540, 3.260~
## $ ef_regulation_business_licensing   <dbl> 7.324582, 8.523503, 8.096776, 5.253~
## $ ef_regulation_business_compliance  <dbl> 7.074366, 7.029528, 6.782923, 6.508~
## $ ef_regulation_business             <dbl> 6.705863, 5.676956, 4.930271, 5.535~
## $ ef_regulation                      <dbl> 6.906901, 5.268992, 5.518500, 5.369~
## $ ef_score                           <dbl> 7.54, 4.99, 5.17, 4.84, 7.57, 7.98,~
## $ ef_rank                            <dbl> 34, 159, 155, 160, 29, 10, 27, 106,~
## $ hf_score                           <dbl> 7.568140, 5.135886, 5.640662, 6.469~
## $ hf_rank                            <dbl> 48, 155, 142, 107, 57, 4, 16, 130, ~
## $ hf_quartile                        <dbl> 2, 4, 4, 3, 2, 1, 1, 4, 2, 2, 4, 2,~
```

Insert your answer here: the dimensions of the dataset are: 1458 observations and 123 variables

2. What type of plot would you use to display the relationship between the personal freedom score, `pf_score`, and one of the other numerical variables? Plot this relationship using the variable `pf_expression_control` as the predictor. Does the relationship look linear? If you knew a country's `pf_expression_control`, or its score out of 10, with 0 being the most, of political pressures and controls on media content, would you be comfortable using a linear model to predict the personal freedom score?

```r
ggplot(hfi, aes(x = pf_expression_control , y = pf_score)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", col = "red", se = TRUE) +
  labs(
    title = "Relationship between pf_score and pf_expression_control"
  )
```



Insert your answer here: A scatterplot is the best plot to display the relationship between two numerical variables. The relationship between pf_expression_control and pf_score appears to be positive and roughly linear, although there is some scatter. Because the pattern is approximately linear, I would be reasonably comfortable using a linear model to predict pf_score from pf_expression_control, though I would still be cautious about variability and possible outliers.

If the relationship looks linear, we can quantify the strength of the relationship with the correlation coefficient.

```
hfi %>%
  summarise(cor(pf_expression_control, pf_score, use = "complete.obs"))
```

```
## # A tibble: 1 x 1
##   `cor(pf_expression_control, pf_score, use = "complete.obs")`
##                                                          <dbl>
## 1                                                        0.796
```

Here, we set the `use` argument to "complete.obs" since there are some observations of NA.

Sum of squared residuals

In this section, you will use an interactive function to investigate what we mean by "sum of squared residuals". You will need to run this function in your console, not in your markdown document. Running the function also requires that the `hfi` dataset is loaded in your environment.

Think back to the way that we described the distribution of a single variable. Recall that we discussed characteristics such as center, spread, and shape. It's also useful to be able to describe the relationship of two numerical variables, such as `pf_expression_control` and `pf_score` above.

3. Looking at your plot from the previous exercise, describe the relationship between these two variables. Make sure to discuss the form, direction, and strength of the relationship as well as any unusual observations.

Insert your answer here: The relationship between pf_expression_control and pf_score appears to be roughly linear, positive, and moderately strong. As pf_expression_control increases, pf_score also tends to increase. There is some scatter around the trend line, so the relationship is not perfect. There may also be a few unusual observations or countries that do not follow the general pattern exactly.

Just as you've used the mean and standard deviation to summarize a single variable, you can summarize the relationship between these two variables by finding the line that best follows their association. Use the following interactive function to select the line that you think does the best job of going through the cloud of points.

```
# This will only work interactively (i.e. will not show in the knitted document)
hfi <- hfi %>% filter(complete.cases(pf_expression_control, pf_score))
DATA606::plot_ss(x = hfi$pf_expression_control, y = hfi$pf_score)
```

After running this command, you'll be prompted to click two points on the plot to define a line. Once you've done that, the line you specified will be shown in black and the residuals in blue. Note that there are 30 residuals, one for each of the 30 observations. Recall that the residuals are the difference between the observed values and the values predicted by the line:

$$e_i = y_i - \hat{y}_i$$

The most common way to do linear regression is to select the line that minimizes the sum of squared residuals. To visualize the squared residuals, you can rerun the plot command and add the argument `showSquares = TRUE`.

```
DATA606::plot_ss(x = hfi$pf_expression_control, y = hfi$pf_score, showSquares = TRUE)
```

Note that the output from the `plot_ss` function provides you with the slope and intercept of your line as well as the sum of squares.

4. Using `plot_ss`, choose a line that does a good job of minimizing the sum of squares. Run the function several times. What was the smallest sum of squares that you got? How does it compare to your neighbors?

Insert your answer here: After running plot_ss several times, the smallest sum of squared residuals I obtained was 1073. My result was different from my neighbor's result, which is expected because each person may click slightly different points when choosing the line.m1

The linear model

It is rather cumbersome to try to get the correct least squares line, i.e. the line that minimizes the sum of squared residuals, through trial and error. Instead, you can use the `lm` function in R to fit the linear model (a.k.a. regression line).

```r
m1 <- lm(pf_score ~ pf_expression_control, data = hfi)
```

The first argument in the function `lm` is a formula that takes the form `y ~ x`. Here it can be read that we want to make a linear model of `pf_score` as a function of `pf_expression_control`. The second argument specifies that R should look in the `hfi` data frame to find the two variables.

The output of `lm` is an object that contains all of the information we need about the linear model that was just fit. We can access this information using the summary function.

```r
summary(m1)
```

```
##
## Call:
## lm(formula = pf_score ~ pf_expression_control, data = hfi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.8467 -0.5704  0.1452  0.6066  3.2060
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            4.61707    0.05745   80.36   <2e-16 ***
## pf_expression_control  0.49143    0.01006   48.85   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8318 on 1376 degrees of freedom
##   (80 observations deleted due to missingness)
## Multiple R-squared:  0.6342, Adjusted R-squared:  0.634
## F-statistic:  2386 on 1 and 1376 DF,  p-value: < 2.2e-16
```

Let's consider this output piece by piece. First, the formula used to describe the model is shown at the top. After the formula you find the five-number summary of the residuals. The "Coefficients" table shown next is key; its first column displays the linear model's y-intercept and the coefficient of `pf_expression_control`. With this table, we can write down the least squares regression line for the linear model:

$$\hat{y} = 4.61707 + 0.49143 \times pf_expression_control$$

One last piece of information we will discuss from the summary output is the Multiple R-squared, or more simply, R^2. The R^2 value represents the proportion of variability in the response variable that is explained by the explanatory variable. For this model, 63.42% of the variability in `pf_free` is explained by `pf_expression_control`.

5. Fit a new model that uses `pf_expression_control` to predict `hf_score`, or the total human freedom score. Using the estimates from the R output, write the equation of the regression line. What does the slope tell us in the context of the relationship between human freedom and the amount of political pressure on media content?

```r
m2 <- lm(hf_score ~ pf_expression_control, data = hfi)
summary(m2)
```

```
##
## Call:
## lm(formula = hf_score ~ pf_expression_control, data = hfi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.6198 -0.4908  0.1031  0.4703  2.2933
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)           5.153687   0.046070  111.87   <2e-16 ***
## pf_expression_control 0.349862   0.008067   43.37   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.667 on 1376 degrees of freedom
##   (80 observations deleted due to missingness)
## Multiple R-squared:  0.5775, Adjusted R-squared:  0.5772
## F-statistic:  1881 on 1 and 1376 DF,  p-value: < 2.2e-16
```

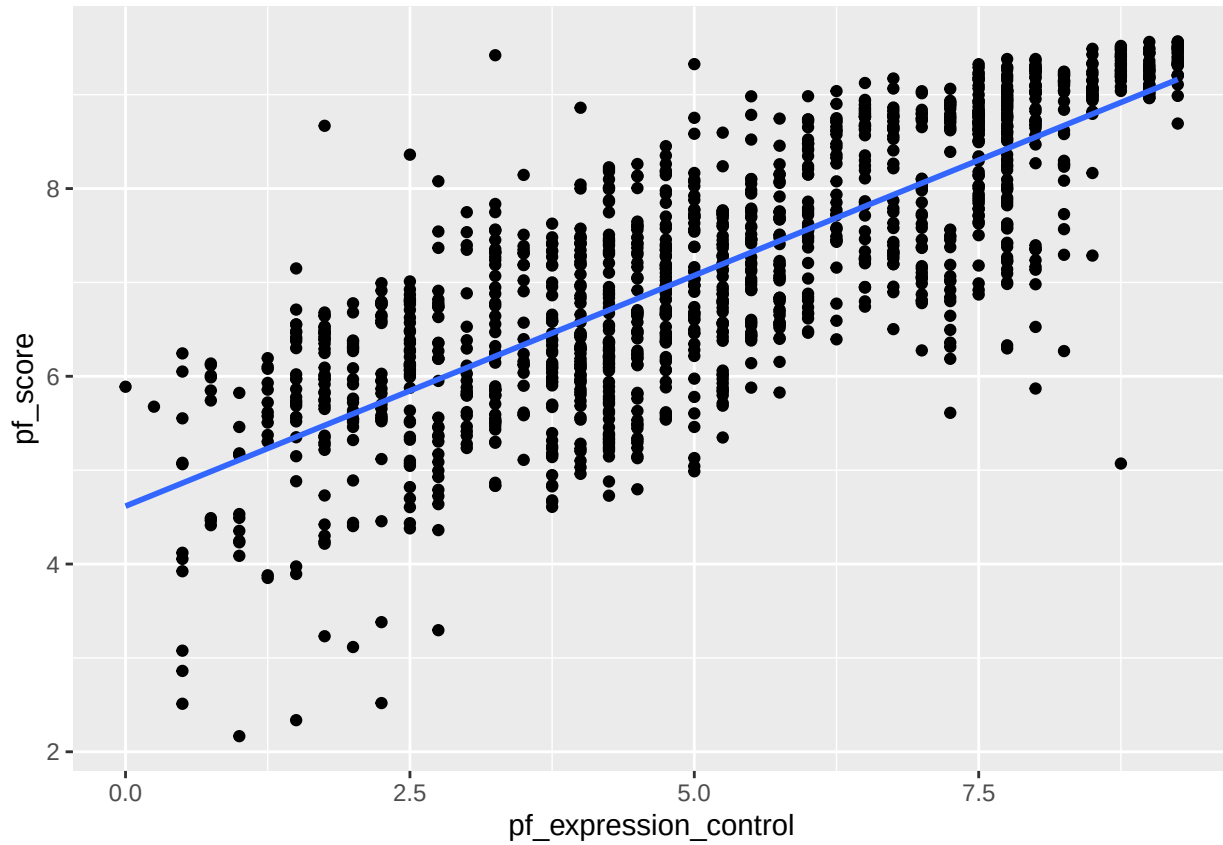
Insert your answer here: What does the slop tell us? The slope indicates that for each 1-point increase in pf_expression_control, the human freedom score increases by approximately 0.35, on average.

$$\hat{y} = 5.153687 + 0.349862 \times pf_expression_control$$

Prediction and prediction errors

Let's create a scatterplot with the least squares line for `m1` laid on top.

```r
ggplot(data = hfi, aes(x = pf_expression_control, y = pf_score)) +
  geom_point() +
  stat_smooth(method = "lm", se = FALSE)
```



Here, we are literally adding a layer on top of our plot. `geom_smooth` creates the line by fitting a linear model. It can also show us the standard error `se` associated with our line, but we'll suppress that for now.

This line can be used to predict y at any value of x. When predictions are made for values of x that are beyond the range of the observed data, it is referred to as *extrapolation* and is not usually recommended. However, predictions made within the range of the data are more reliable. They're also used to compute the residuals.

6. If someone saw the least squares regression line and not the actual data, how would they predict a country's personal freedom score for one with a 6.7 rating for `pf_expression_control`? Is this an overestimate or an underestimate, and by how much? In other words, what is the residual for this prediction?

Insert your answer here: Using the regression line, the predicted pf_score is approximately 7.91. To determine whether this is an overestimate or underestimate, we would compare it to the actual observed value. The residual is the difference between the actual value and the predicted value. if we use the formula above m1:

$$\text{residual} = y - \hat{y}$$

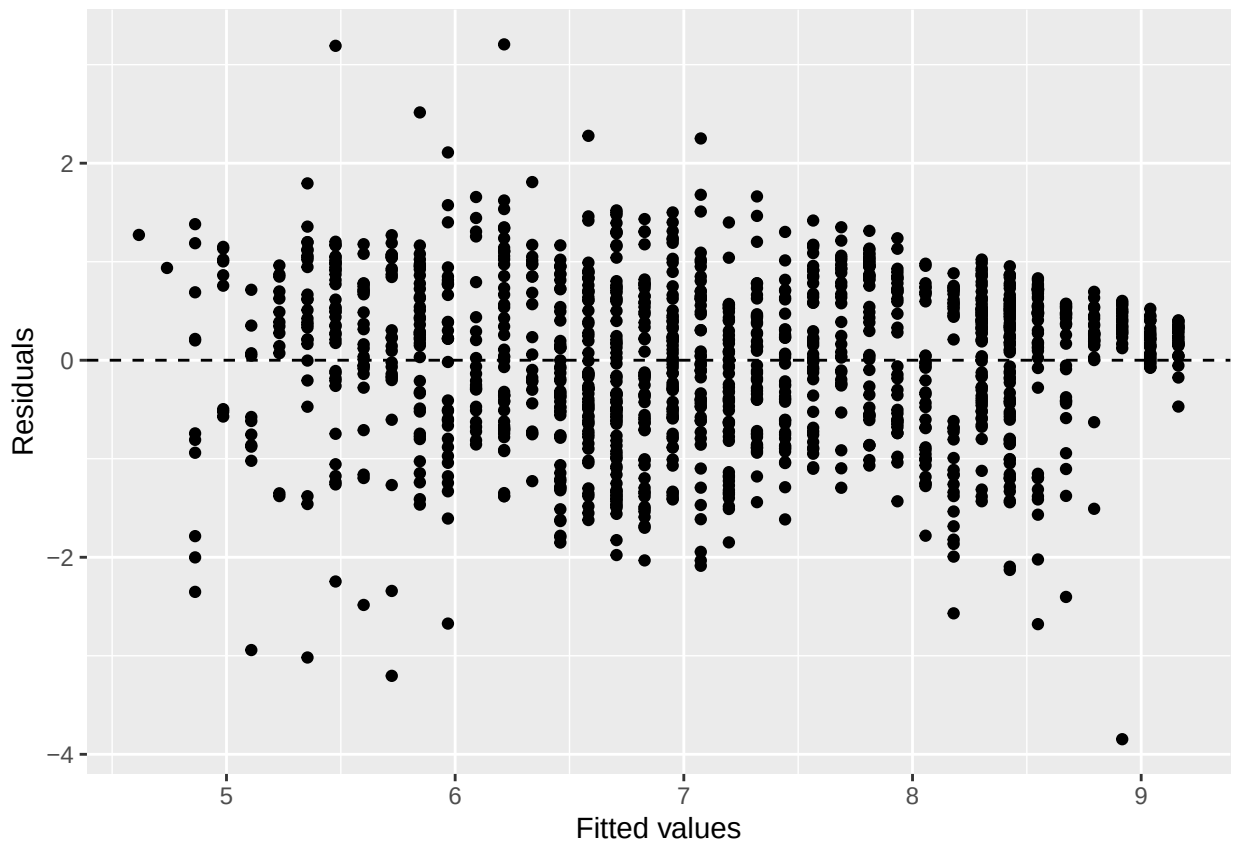
$$\hat{y} = 4.61707 + 0.49143 \times 6.7 \approx 7.91$$

Model diagnostics

To assess whether the linear model is reliable, we need to check for (1) linearity, (2) nearly normal residuals, and (3) constant variability.

Linearity: You already checked if the relationship between `pf_score` and 'pf_expression_control' is linear using a scatterplot. We should also verify this condition with a plot of the residuals vs. fitted (predicted) values.

```
ggplot(data = m1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  xlab("Fitted values") +
  ylab("Residuals")
```



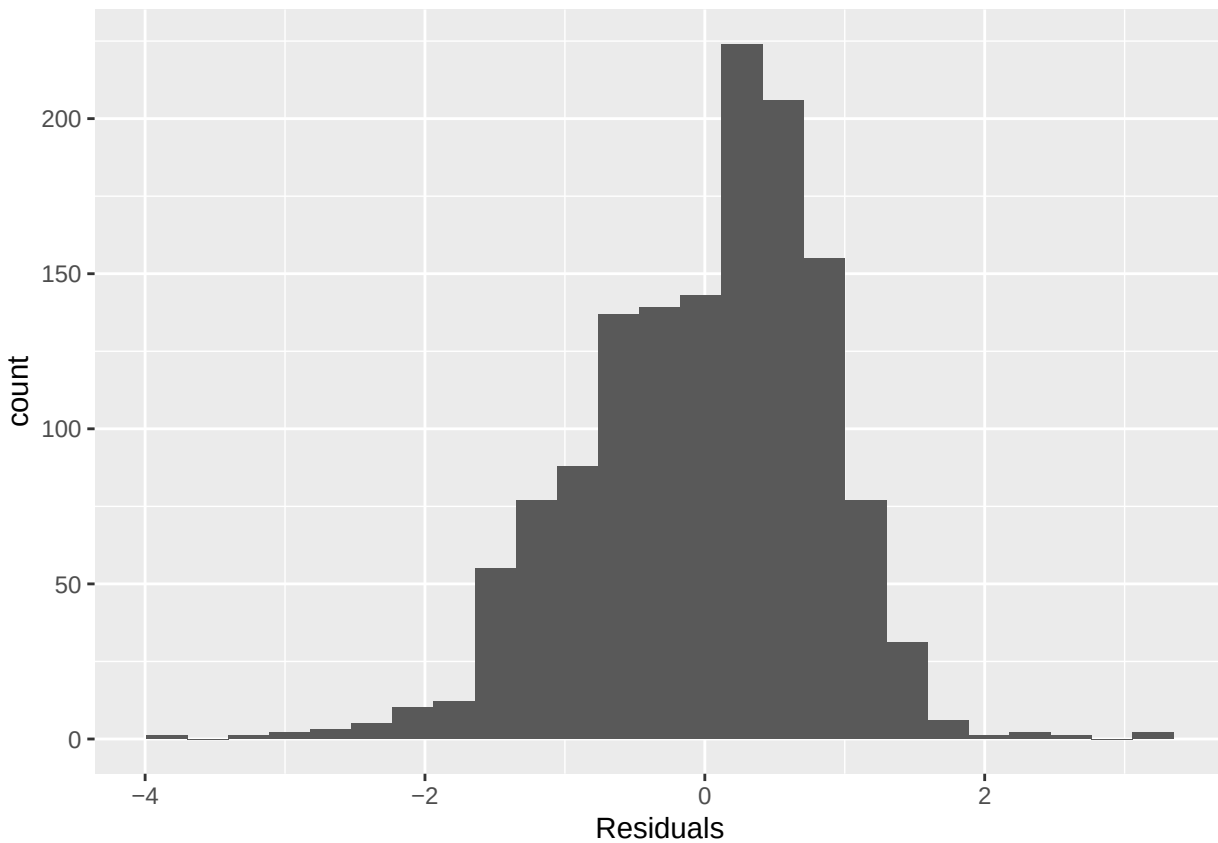
Notice here that `m1` can also serve as a data set because stored within it are the fitted values ($\hat{y}$) and the residuals. Also note that we're getting fancy with the code here. After creating the scatterplot on the first layer (first line of code), we overlay a horizontal dashed line at $y = 0$ (to help us check whether residuals are distributed around 0), and we also reanme the axis labels to be more informative.

7. Is there any apparent pattern in the residuals plot? What does this indicate about the linearity of the relationship between the two variables?

Insert your answer here: There is no clear pattern in the residuals; they appear randomly scattered around zero. This suggests that the linearity assumption is reasonable and that a linear model is appropriate.

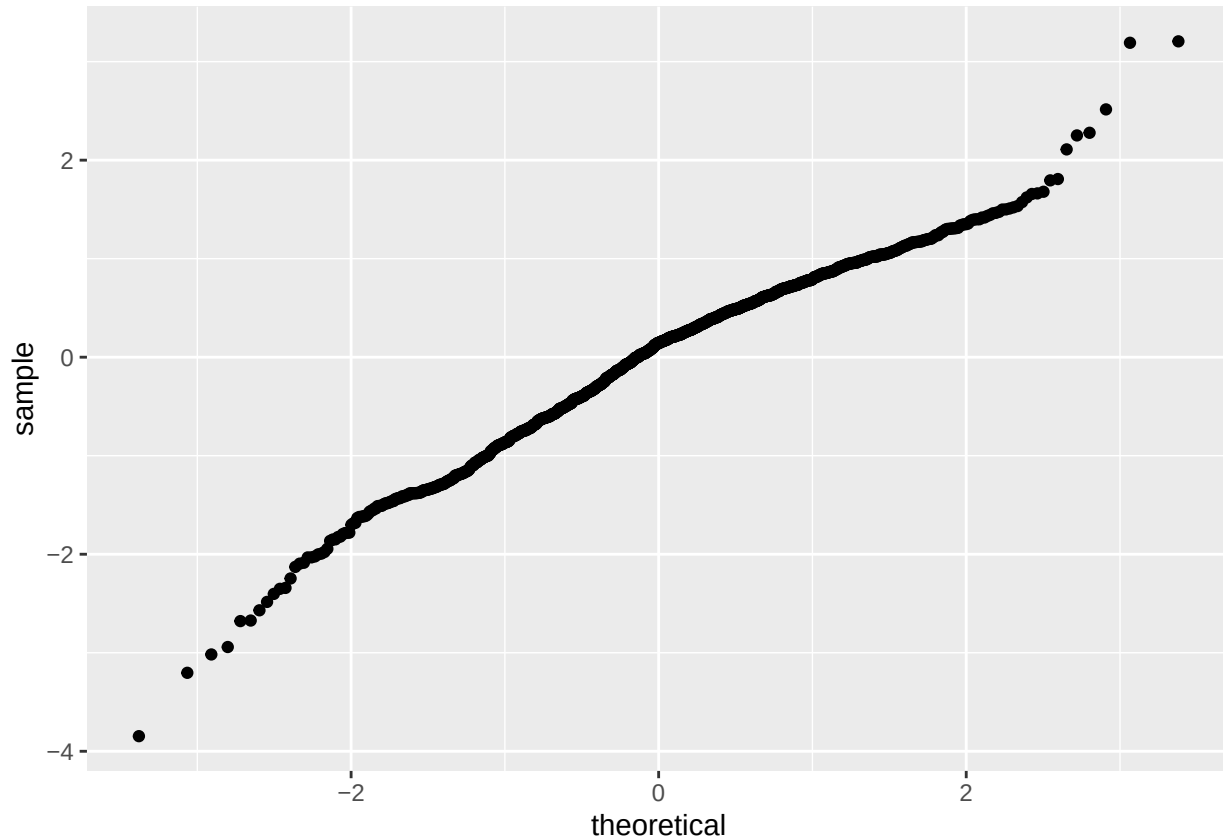
Nearly normal residuals: To check this condition, we can look at a histogram

```r
ggplot(data = m1, aes(x = .resid)) +
  geom_histogram(bins = 25) +
  xlab("Residuals")
```



or a normal probability plot of the residuals.

```r
ggplot(data = m1, aes(sample = .resid)) +
  stat_qq()
```



Note that the syntax for making a normal probability plot is a bit different than what you're used to seeing: we set `sample` equal to the residuals instead of `x`, and we set a statistical method `qq`, which stands for "quantile-quantile", another name commonly used for normal probability plots.

8. Based on the histogram and the normal probability plot, does the nearly normal residuals condition appear to be met?

Insert your answer here: Yes, the nearly normal residuals condition appears to be met. The histogram is approximately symmetric and bell-shaped, and the points in the normal probability (QQ) plot fall roughly along a straight line.

Constant variability:

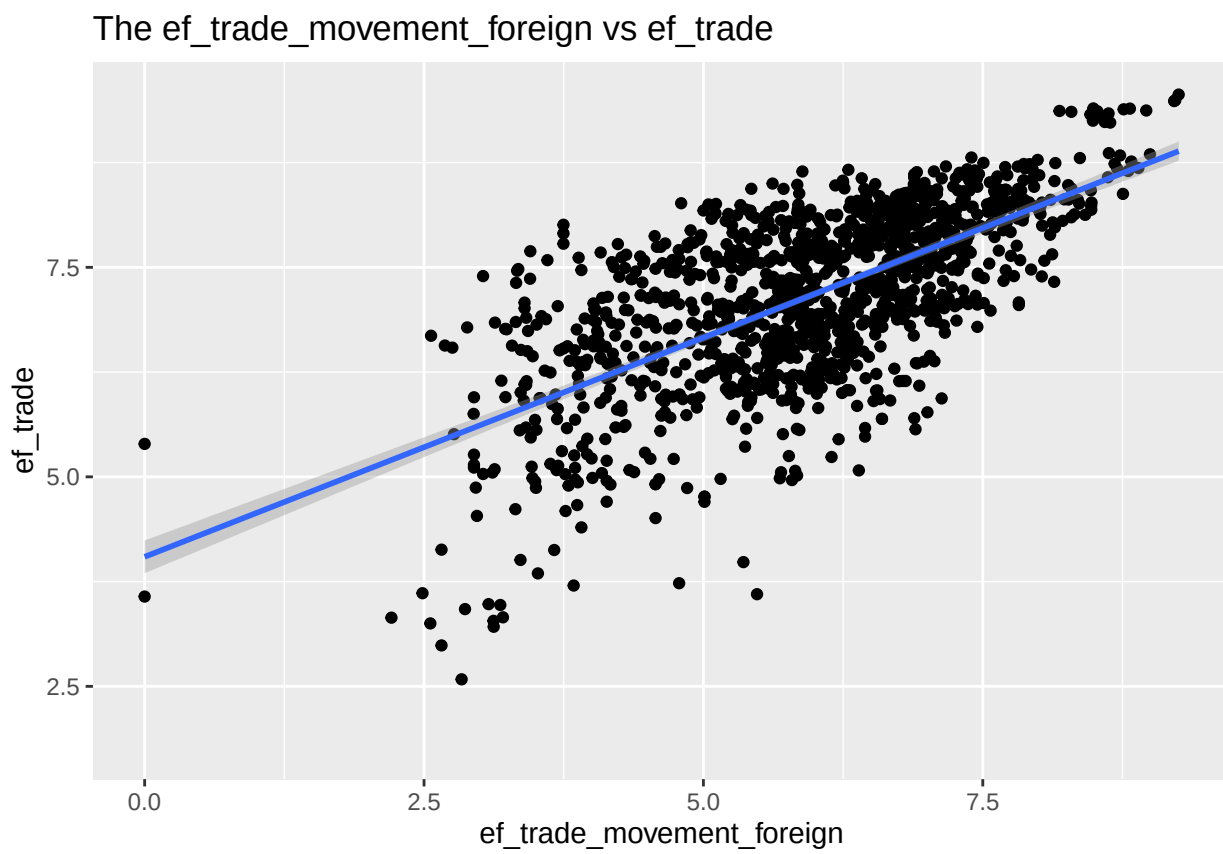
9. Based on the residuals vs. fitted plot, does the constant variability condition appear to be met?

Insert your answer here: Yes, the constant variability condition appears to be met because the residuals show a roughly constant spread across all fitted values, with no clear funnel shape or pattern.

More Practice

- Choose another freedom variable and a variable you think would strongly correlate with it.. Produce a scatterplot of the two variables and fit a linear model. At a glance, does there seem to be a linear relationship?

```r
ggplot( data = hfi, aes(x = ef_trade_movement_foreign, y = ef_trade)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "The ef_trade_movement_foreign vs ef_trade",
    x = "ef_trade_movement_foreign",
    y = "ef_trade"
  )
```



Insert your answer here: Yes, there appears to be a moderately strong positive linear relationship between ef_trade_movement_foreign and ef_trade. As the movement of foreign trade increases, the overall trade freedom score also increases. The points follow a roughly linear pattern, although there is some scatter.

- How does this relationship compare to the relationship between `pf_expression_control` and `pf_score`? Use the R^2 values from the two model summaries to compare. Does your independent variable seem to predict your dependent one better? Why or why not?

```r
m3 <- lm(ef_trade ~ ef_trade_movement_foreign, data = hfi)
summary(m3)
```

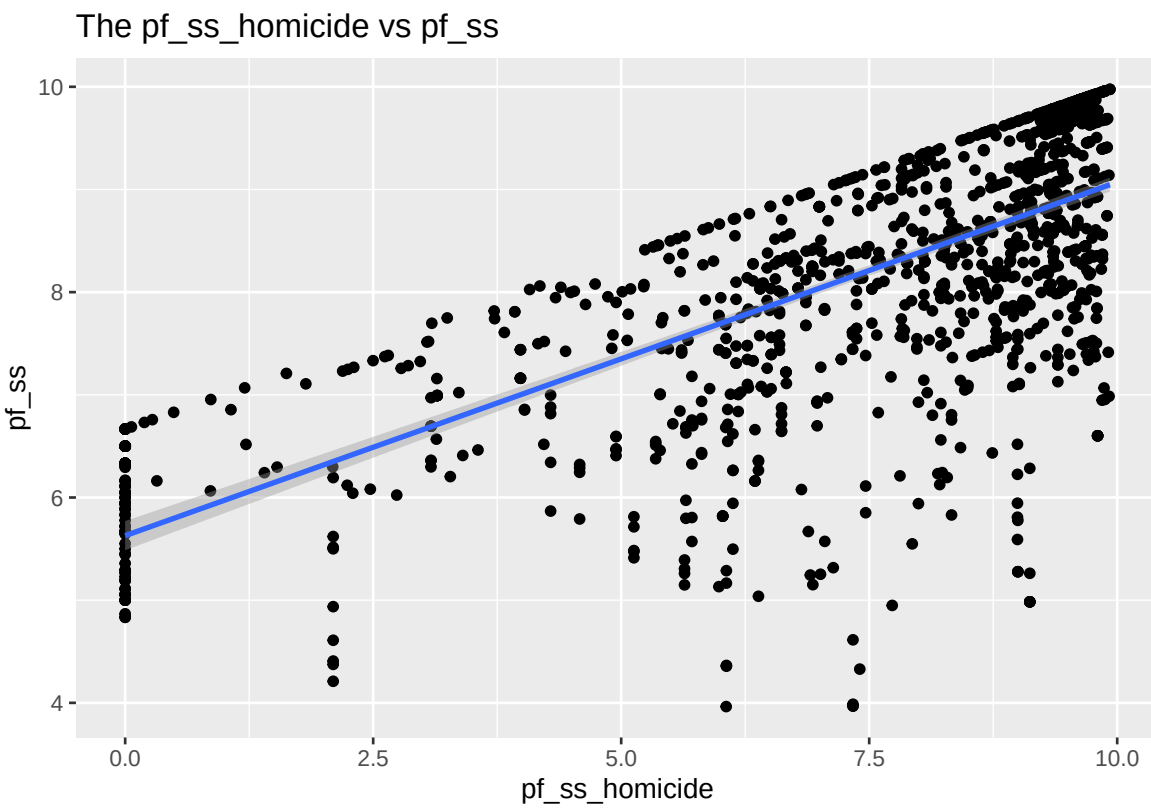
```
##
## Call:
## lm(formula = ef_trade ~ ef_trade_movement_foreign, data = hfi)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.3129 -0.4923  0.0879  0.5569  2.0018
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                4.04560    0.09964   40.60   <2e-16 ***
## ef_trade_movement_foreign  0.52317    0.01640   31.91   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.782 on 1291 degrees of freedom
##   (165 observations deleted due to missingness)
## Multiple R-squared:  0.4409, Adjusted R-squared:  0.4405
## F-statistic:  1018 on 1 and 1291 DF,  p-value: < 2.2e-16
```

Insert your answer here: The R^2 value for the ef_trade model is about 0.44, meaning that approximately 44% of the variability in ef_trade is explained by ef_trade_movement_foreign. This is lower than the R^2 value for the relationship between pf_expression_control and pf_score (about 63%), which indicates that pf_expression_control is a stronger predictor. Therefore, the earlier model does a better job explaining variability in the response variable.

- What's one freedom relationship you were most surprised about and why? Display the model diagnostics for the regression model analyzing this relationship.

```r
ggplot( data = hfi, aes(x = pf_ss_homicide, y = pf_ss)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
title = "The pf_ss_homicide vs pf_ss",
x = "pf_ss_homicide",
y = "pf_ss"
  )
```

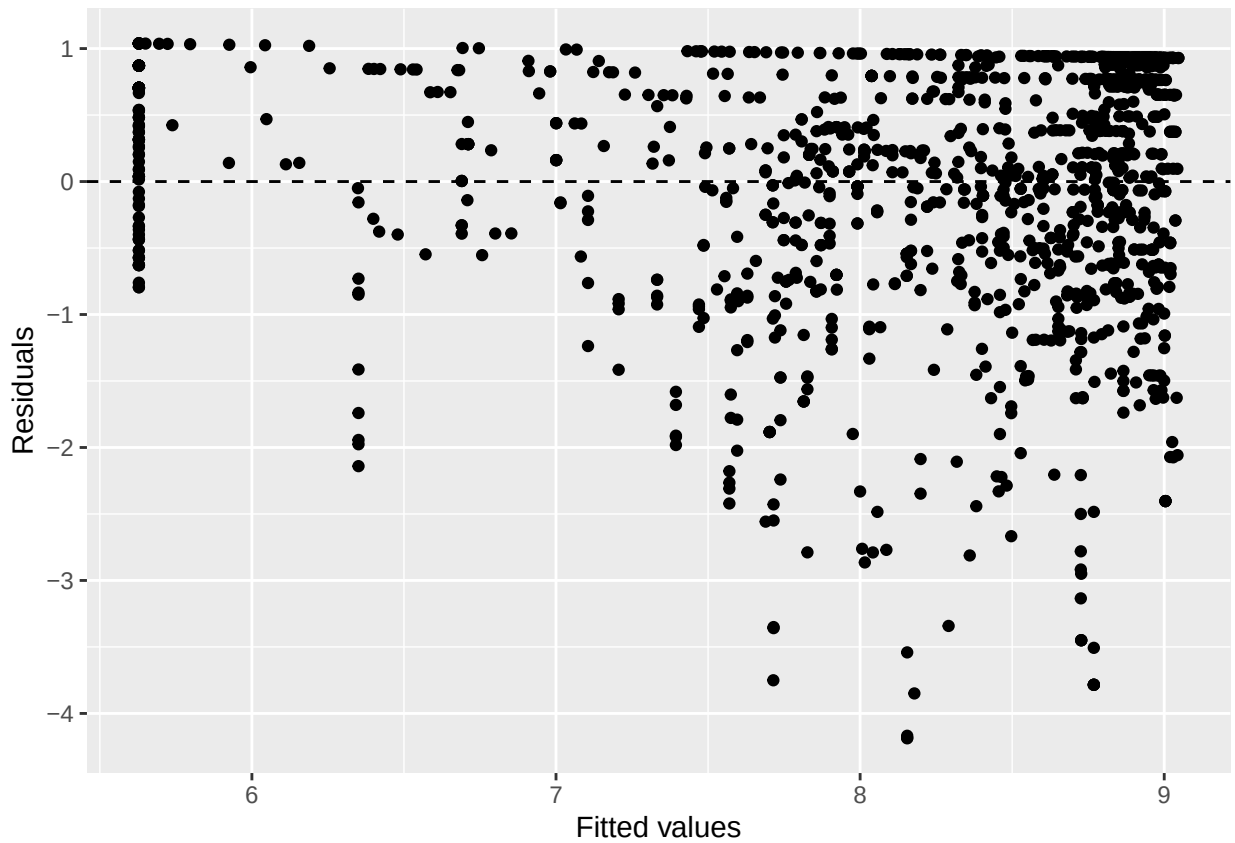


Insert your answer here: I was surprised by the relationship between pf_ss_homicide and pf_ss. The scatterplot shows a positive linear trend, and the regression model confirms a moderate relationship with an R^2 of about 0.51. This indicates that homicide-related safety is strongly associated with overall personal safety and security. The linear model appears to be reasonably appropriate, although there is still some variability in the data

```r
m4 <- lm(pf_ss ~ pf_ss_homicide, data = hfi)
summary(m4)
```

```
##
## Call:
## lm(formula = pf_ss ~ pf_ss_homicide, data = hfi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.1870 -0.5256  0.2066  0.7920  1.0382
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.628455   0.072489   77.64   <2e-16 ***
## pf_ss_homicide 0.344346   0.009135   37.70   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9603 on 1376 degrees of freedom
##   (80 observations deleted due to missingness)
## Multiple R-squared:  0.508,  Adjusted R-squared:  0.5077
## F-statistic:  1421 on 1 and 1376 DF,  p-value: < 2.2e-16
```

```r
ggplot(data = m4, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  xlab("Fitted values") +
  ylab("Residuals")
```



Interpretation: The residuals appear to be randomly scattered around zero, suggesting that the linearity assumption is reasonable. There is no strong pattern indicating non-linearity. The spread of residuals is relatively constant, so the constant variability assumption appears to be met. Overall, the linear model seems appropriate for this relationship. * * *